

Long-Run Economic Growth: Sources and Policies

ECON 101 - Macroeconomics

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Winter 2026

- **7.1 The Importance of Economic Growth**
- 7.2 The Economic Growth Model
- 7.3 New Growth Theory and Knowledge Capital
- 7.4 Why Isn't the Whole World Rich?

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Obtaining Economic Growth

- Previous chapter: how to **measure** growth (real GDP, real GDP per capita).
- This chapter: how to **explain** and **influence** long-run growth.
- Key idea:
 - Economic growth is **not inevitable**.
 - History includes long periods of stagnation with little or no increase in output per person.
- Central questions:
 - Why have some countries achieved rapid, sustained growth?
 - Why have others failed to keep up and remained poor?

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- Application of mechanical power to the production of goods and services.
- Began in England around 1750.

- Before that:

- Production relied mainly on human and animal muscle power.
- Very slow (almost zero) growth in real GDP per capita.

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- England and later the United States, France, Germany, and others
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Table 7.1 – The Effects of Different Growth Rates on Living Standards (Placeholder)

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Instruction: Reproduce the exact table from the textbook here, with the same headings and numerical values showing how different constant growth rates (e.g., 1%, 2%, 3%) change income levels over time (e.g., 35, 70 years).

Insert figure/diagram here later.

Why Do Growth Rates Matter?

- Small differences in annual growth rates lead to **huge** differences in living standards over decades.
- A country that grows too slowly fails to raise living standards:
 - Not just about smartphones and TVs,
 - but about basic indicators:
 - infant mortality,
 - life expectancy,
 - access to clean water, health care, and education.
- Example contrast:
 - In many high-income countries, fewer than 6 of 1,000 babies die before age one.
 - In some of the poorest countries, more than 50 of 1,000 die before age one.

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Differences in Incomes Across Countries

- Economists often divide economies into:
 - **High-income (industrial) countries:**
 - e.g., Australia, Canada, Japan, New Zealand, the United States, Western Europe.
 - **Developing (low- and middle-income) countries:**
 - many countries in Africa, Asia, and Latin America.
- Some economies, like Singapore, South Korea, and Taiwan, have become **newly industrializing countries** with very high growth rates.
- Real GDP per capita differs dramatically even after adjusting for cost-of-living differences.
- Example from 2020:
 - very high income: Monaco (over \$170,000 per person),
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Figure 7.1 – Real GDP per Capita Around the World (Diagram Placeholder)

Figure 7.1 – Map or bar chart of real GDP per capita across countries, showing large gaps between high-income, newly industrializing, and developing countries.

Insert figure/diagram here later.

7.2 Learning Objective

Goal

Use the economic growth model to explain why growth rates differ across countries.

- An **economic growth model** explains growth rates in **real GDP per capita** over the long run.
- Key idea from Chapter 6:
 - Long-run growth in real GDP per person is driven by growth in labour productivity.

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Labour Productivity and Its Determinants

- **Labour productivity:**

$$\text{Labour productivity} = \frac{\text{Quantity of goods and services produced}}{\text{Number of workers or hours worked}}.$$

- Two main factors determine labour productivity:

1. Quantity of capital available to workers
2. Level of technology

- Our model focuses on:

- changes in the quantity of capital per worker, and
- **technological change:** a change in the amount of output a firm can produce with a given quantity of inputs.

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- These allow each worker to produce **more** per hour with the same effort.
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 - and experience.
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 - workers can adopt new technologies more quickly,
 - they can solve problems more effectively,
 - they are more flexible when the economy changes.

- Sometimes productivity rises not because of new machines, but because of:
 - better management,
 - improved organization of production,
 - more efficient supply chains.
- Example:
 - Just-in-time inventory systems reduce waste and allow firms to respond quickly to demand, raising output per worker.

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The Per-Worker Production Function

- To formalize these ideas, we use a **per-worker production function**:
 - It shows the relationship between **real GDP per hour worked** and **capital per hour worked**, holding technology constant.
- With a fixed level of technology:
 - The first units of capital are the most effective.
 - Additional units increase output, but by **smaller and smaller** amounts.
- This is called **diminishing returns** to capital.

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Figure 7.3 – The Per-Worker Production Function (Diagram Placeholder)

Figure 7.3 – Per-worker production function: real GDP per hour worked vs. capital per hour worked, with diminishing returns.

Insert figure/diagram here later.

- **Diminishing returns:**

- Increasing one input (capital) progressively, while holding other inputs constant, yields smaller and smaller increases in output.

- Intuition:

- Give a worker their first machine or computer: huge productivity gain.
- Give them the tenth very similar machine: smaller extra gain.

- This means that increasing capital per worker alone cannot generate high growth forever.

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Technological Change and the Production Function

- When technology improves, the entire per-worker production function shifts **upward**.
- This means:
 - For any given amount of capital per worker,
 - output per worker is higher.
- Technological change is therefore crucial for sustained long-run growth.

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Figure 7.4 – Technological Change Increases Output (1 of 2) (Diagram Placeholder)

Figure 7.4 (1 of 2) – For a country with low capital per worker, increases in capital are very effective at raising real GDP per hour worked.

Insert figure/diagram here later.

Figure 7.4 – Technological Change Increases Output (2 of 2) (Diagram Placeholder)

Figure 7.4 (2 of 2) – With high capital per worker, further increases in capital face strong diminishing returns; technological change becomes the main source of growth.

Insert figure/diagram here later.

Capital Deepening vs. Technological Change

- **Capital deepening:**

- An increase in the amount of capital per worker.
- In relatively poor countries:
 - Capital deepening can produce large gains in productivity.
- In relatively rich countries:
 - Capital deepening still helps, but diminishing returns set in.
 - Continuing growth in living standards requires **sustained technological change**.

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- In the Solow model:
 - Productivity growth is key to long-run growth.
 - But technological change is treated as something that happens **outside** the model (an exogenous factor).
- **New growth theory**, developed by **Paul Romer** and others:
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- Knowledge capital:
 - The stock of ideas, scientific knowledge, designs, and blueprints.
 - Generated by research and development (R&D) and other innovative activities.
- Increases in knowledge capital arise from:
 - R&D in firms and universities,
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Physical Capital vs. Knowledge Capital

- **Physical capital:**

- Rival and excludable (a private good).
- One machine can only be used by one worker at a time.
- Leads to **diminishing returns**.

- **Knowledge capital:**

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- One firm's use of an idea does not prevent others from using it.
- At the level of the whole economy, this can generate **increasing returns to scale**.

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Free-Rider Problem and Underinvestment in Knowledge

- Because knowledge capital is a public good, firms may not capture the full benefits of their R&D.
- **Free-rider problem:**
 - People or firms can benefit from an idea without paying for it.
- **Example:**
 - A major lab develops a transistor or a new drug.
 - Other firms learn from this and earn profits, even though they did not bear the original research cost.
- **Result:**
 - Market economy produces **too little** knowledge capital relative to the socially optimal amount.

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- Allow firms to earn profits from their inventions, increasing their incentive to innovate.

- **Copyrights:**

- Similar protection for creative works (books, films, music).
- Typically last for the creator's life plus many years after.

- **Policy trade-off:**

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- They subsidize R&D at universities and private firms through grants and tax incentives.

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 - High-quality education helps create a workforce capable of doing research and adopting new technologies.
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7.4 Learning Objective

Goal

Explain economic catch-up and why many poor countries have not experienced rapid growth.

- The economic growth model predicts that **poor countries should grow faster** than rich ones.
- This would imply that poor countries eventually **catch up** to the income levels of rich countries.

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- **Catch-up (convergence):**

- Prediction that GDP per capita in poor countries will grow faster than in rich countries.
- Why might we expect this?
 - Additional capital is more productive in countries with small capital stocks.
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Figure 7.6 – Catch-Up Predicted by the Model (Diagram Placeholder)

Figure 7.6 – Hypothetical graph showing real GDP per capita in poor and rich countries converging over time as poor countries grow faster.

Insert figure/diagram here later.

Evidence of Catch-Up Among High-Income Countries

- When we look only at high-income countries, we see evidence of catch-up:
 - Countries that were relatively poor in 1960 (e.g., Hong Kong, South Korea) had faster growth.
 - Countries that were relatively rich in 1960 (e.g., the United States, Switzerland) had slower growth.
- So among high-income countries, convergence appears to have occurred over the postwar period.

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Figure 7.7 – Catch-Up Among High-Income Countries (Diagram Placeholder)

Figure 7.7 – Growth from 1960 to recent years plotted against initial real GDP per capita for high-income countries, showing a negative relationship (poorer grew faster).

Insert figure/diagram here later.

Most of the World Has Not Been Catching Up

- When we extend the analysis to **all** countries, the simple catch-up prediction performs poorly.
- Many low-income countries:
 - have grown much more slowly than high-income countries,
 - and some have even experienced periods of negative growth.
- So we need to understand:
 - Why some poor countries grow quickly and others do not.

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Figure 7.8 – Most of the World Has Not Been Catching Up (Diagram Placeholder)

Figure 7.8 – Scatter plot of growth vs. initial income for many countries, showing weak or no convergence once low-income countries are included.

Insert figure/diagram here later.

Why Have High-Income Countries Stopped Catching Up to the U.S.?

- In 1990, many high-income countries had real GDP per capita somewhat below the U.S. level.
- By 2019, many had **not** closed the gap and some had fallen further behind.
- Possible reasons:
 - U.S. labour markets are relatively **flexible**.
 - U.S. workers tend to:
 - enter the labour force earlier,
 - retire later.
 - The U.S. financial system is relatively **efficient** and **liquid**, making it attractive for investment.
 - Access to venture capital and support for start-ups is strong, helping new firms grow.

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Figure 7.9 – Other High-Income Countries vs. the U.S. (Diagram Placeholder)

Figure 7.9 – Bars showing real GDP per capita relative to the U.S. in 1990 and in 2019 for selected high-income countries; most red bars are equal or lower than blue bars.

Insert figure/diagram here later.

Why Are Many Poor Countries Growing So Slowly?

- Economists often highlight four key factors:
 1. Failure to enforce the rule of law
 2. Wars and revolutions
 3. Poor public education and health
 4. Low rates of saving and investment

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- **Rule of law:**

- Government's ability to enforce the laws, particularly with respect to:
 - protecting private property,
 - enforcing contracts.

- **Property rights:**

- Individuals and firms must have the exclusive right to use their property and to buy or sell it.

- **Without strong rule of law:**

- entrepreneurs fear expropriation and corruption,
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- Wars and internal conflicts:
 - destroy physical capital,
 - disrupt education and health systems,
 - create uncertainty that discourages investment.
- Example:
 - A country experiencing civil war may have the same real GDP per capita decades apart.
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- Weak public schools and limited access to education:
 - reduce **human capital**,
 - make it harder for workers to adopt new technologies.
- Poor health care and inadequate nutrition:
 - lower worker productivity,
 - reduce the number of hours people are able to work.
- Investing in education and health is both:
 - a social policy,
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Low Rates of Saving and Investment

- Many low-income countries have:
 - underdeveloped and insecure financial systems,
 - limited access to credit for households and firms.
- This creates a **vicious cycle**:
 - Low income → low saving → low investment → slow growth → low income.
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Summary

- Long-run economic growth is crucial for improving living standards.
- Growth models emphasize:
 - labour productivity,
 - capital per worker,
 - and technological change.
- New growth theory places **knowledge capital** and **incentives** at the centre of growth.
- Governments can promote growth by:
 - protecting intellectual property,
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- While some countries have caught up to high-income levels, many have not, due to:
 - weak rule of law,
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Thank you!